EXERCISE 02

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1 Introduction

Solution of Exercise 02 on ranking binary numbers with a perceptron.

2 Methodology

2.1 Benchmark

The benchmark converts the binary numbers (encoded using ± 1) into decimal integers and compares them directly.
It is implemented independently of the perfect perceptron, making it a good benchmark.

2.2 Training

A perfect perceptron supplies the correct labels. The student weights are initialized from $\mathcal{N}(0, 1)$ and updated only after an incorrect prediction, using

$$\mathbf{J} \leftarrow \mathbf{J} + \frac{1}{\sqrt{20}}\, \sigma_T^\mu \boldsymbol{\xi}^\mu. \tag{1}$$

The training examples are presented sequentially in a fixed order. After each complete cycle (epoch), the training error is evaluated. The procedure is repeated until the training error reaches exactly zero.

2.3 Performance measures

Training and test performance are measured by the fraction of misclassified examples,

$$\epsilon = \frac{1}{P} \sum_{\mu=1}^{P} \mathbf{1}[\sigma^\mu \neq \sigma_T^\mu]\,. \tag{2}$$

The test error is evaluated on 10^4 independently generated examples.

3 Results

3.1 Classification errors

Table 1: Training and test errors.

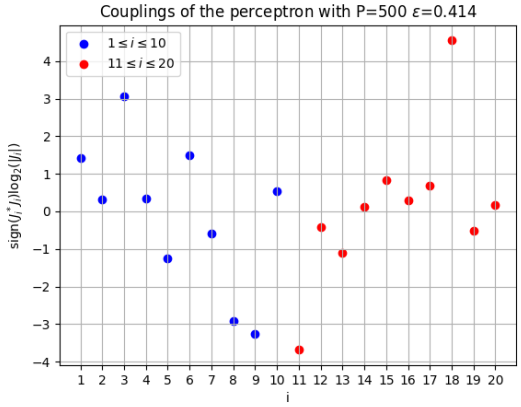
Perceptron	Training error	Test error
Perfect (teacher)	—	0
Random (untrained)	—	0.5107
Trained, $P = 500$	0	0.0168
Trained, $P = 2000$	0	0.0032

The random perceptron has random couplings drawn from $\mathcal{N}(0, 1)$. Since these couplings contain no information about the teacher, its expected classification error is 0.5, consistent with the observed value of 0.5107. An error close to zero indicates successful learning, whereas an error close to one indicates that the student systematically predicts the opposite of the teacher, and therefore still contains information about the hidden rule.
Both trained perceptrons reach zero training error. Increasing P from 500 to 2000 reduces the observed test error from 1.68% to 0.32%.

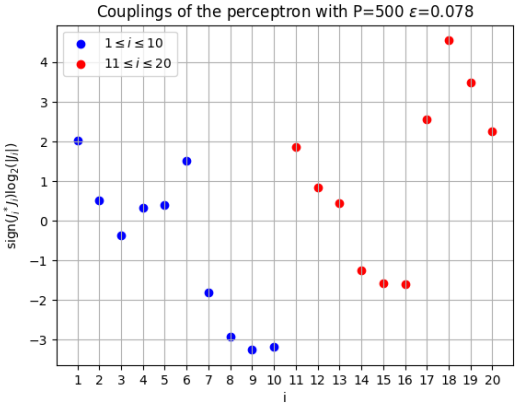
3.2 Evolution of the couplings

The profiles show $q_i = \text{sign}(J_i^* J_i) \log_2 |J_i|$ at four stages of training.
We can see that perfect classification during training does not require the student's weights to reproduce the teacher's weights for every index.

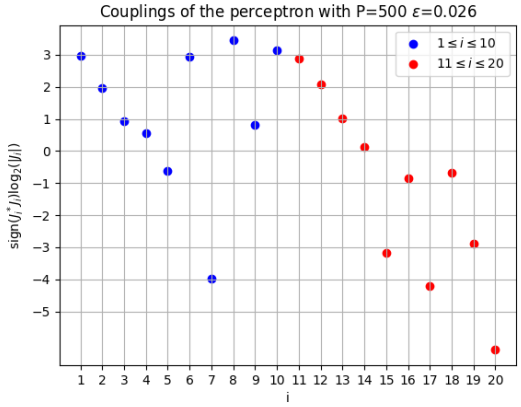
3.2.1 $P = 500$



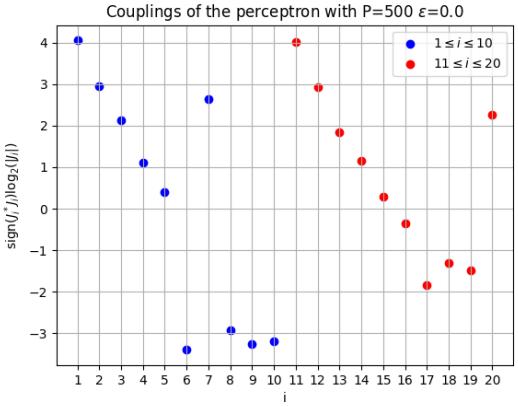
(a) Initial: $\epsilon_t = 0.414$



(b) Intermediate: $\epsilon_t = 0.078$



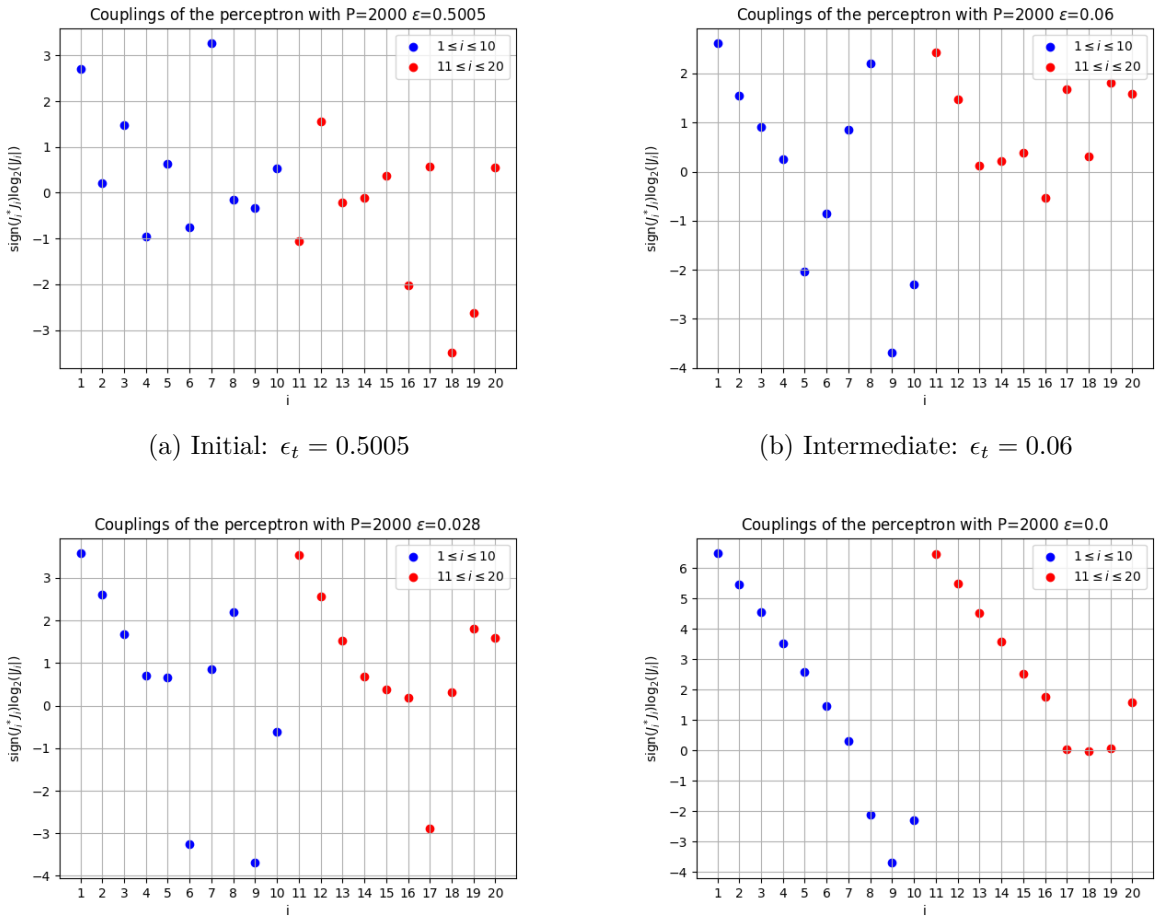
(c) Late: $\epsilon_t = 0.026$



(d) Final: $\epsilon_t = 0$

Figure 1: Coupling profiles during training with 500 examples.

3.2.2 $P = 2000$



(a) Initial: $\epsilon_t = 0.5005$

(b) Intermediate: $\epsilon_t = 0.06$

(c) Late: $\epsilon_t = 0.028$

(d) Final: $\epsilon_t = 0$

Figure 2: Coupling profiles during training with 2000 examples.

An interesting feature, observed across different random seeds, is that the training error initially decreases rapidly, whereas convergence becomes considerably slower as the error approaches zero. In several runs, reducing the error from approximately 0.02 to zero required more training than reaching 0.02 from the initial random configuration. This effect was particularly pronounced for $P = 2000$, which generally required more training epochs than $P = 500$.

4 Conclusions

The perceptron trained on 2000 examples achieves a lower generalization error than the one trained on 500 examples, although both reach zero training error.
The learned couplings need not coincide with those of the teacher.
Furthermore, convergence becomes considerably slower near zero training error, particularly for the larger dataset.